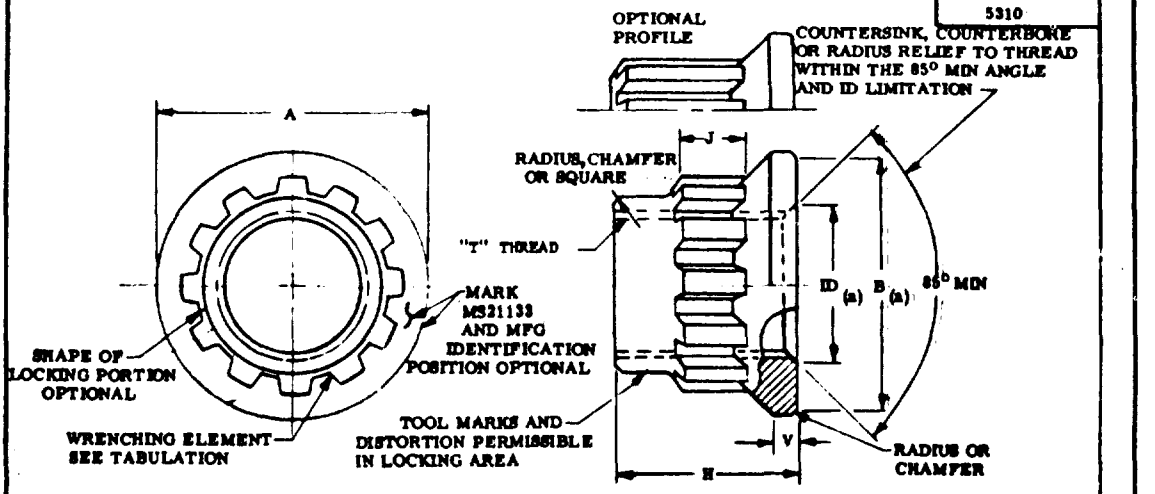
FED. SUP CLASS
5310




| NOMINAL SIZE | DASH NUMBER NON-DRY LUB | DASH NUMBER DRY LUB | "T" THREAD | MS33787 ELEMENT NUMBER | A DIA MAX | B DIA MIN (a) | ID (a) MAX | ID (a) MIN | H MAX | V MIN | J MIN | X | ULTIMATE AXIAL STRENGTH LB MIN (b) | APPROX WEIGHT LB/100 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NO.10 | 03 | L03 | .1900-32UNJF-3B | 8 | .346 | .306 | .220 | .190 | .190 | .021 | .084 | .003 | 3,910 | .21 |
| 1/4 | 04 | L04 | .2500-28UNJF-3B | 10 | .439 | .399 | .280 | .250 | .250 | .028 | .110 | .003 | 6,980 | .42 |
| 5/16 | 05 | L05 | .3125-24UNJF-3B | 12 | .534 | .494 | .342 | .312 | .312 | .034 | .138 | .004 | 11,100 | .72 |
| 3/8 | 06 | L06 | .3750-24UNJF-3B | 14 | .634 | .594 | .405 | .375 | .375 | .041 | .165 | .004 | 17,100 | 1.16 |
| 7/16 | 07 | L07 | .4375-20UNJF-3B | 16 | .733 | .693 | .473 | .438 | .438 | .048 | .193 | .005 | 23,200 | 1.90 |
| 1/2 | 08 | L08 | .5000-20UNJF-3B | 20 | .833 | .793 | .535 | .500 | .500 | .055 | .220 | .005 | 30,900 | 2.80 |
| 9/16 | 09 | L09 | .5625-18UNJF-3B | 22 | .929 | .889 | .597 | .562 | .562 | .062 | .248 | .005 | 39,200 | 3.80 |
| 5/8 | 10 | L10 | .6250-18UNJF-3B | 24 | 1.030 | .990 | .660 | .625 | .625 | .069 | .275 | .006 | 49,000 | 5.10 |
| 3/4 | 12 | L12 | .7500-16UNJF-3B | 30 | 1.224 | 1.184 | .785 | .750 | .750 | .083 | .330 | .007 | 71,100 | 9.20 |
| 7/8 | 14 | L14 | .8750-14UNJF-3B | 34 | 1.419 | 1.379 | .910 | .875 | .875 | .096 | .385 | .008 | 97,100 | 14.00 |
| 1 | 16 | L16 | 1.0000-12UNJF-3B | 38 | 1.620 | 1.570 | 1.035 | 1.000 | 1.000 | .110 | .440 | .009 | 126,000 | 20.00 |
| 1 1/8 | 18 | L18 | 1.1250-12UNJF-3B | 44 | 1.822 | 1.772 | 1.160 | 1.125 | 1.125 | .124 | .495 | .010 | 162,000 | 29.00 |
| 1 1/4 | 20 | L20 | 1.2500-12UNJF-3B | 46 | 2.022 | 1.972 | 1.285 | 1.250 | 1.250 | .138 | .550 | .011 | 202,000 | 40.00 |
| 1 3/8 | 22 | L22 | 1.3750-12UNJF-3B | 52 | 2.221 | 2.171 | 1.410 | 1.375 | 1.375 | .151 | .605 | .012 | 247,000 | 52.00 |
| 1 1/2 | 24 | L24 | 1.5000-12UNJF-3B | 56 | 2.423 | 2.373 | 1.535 | 1.500 | 1.500 | .165 | .660 | .013 | 296,000 | 67.00 |

(a) MINIMUM BEARING AREA BASED ON A BEARING STRESS OF 115 KSI.
(b) AXIAL STRENGTH DETERMINED FROM FORMULA $W_a = F_{tu}A$ WHERE A IS THE CROSS SECTIONAL AREA, IN SQUARE INCHES, BASED ON THE MAXIMUM PITCH DIAMETER OF BOLT THREAD, $F_{tu}$ IS 180 KSI AND $W_a$ IS THE AXIAL STRENGTH IN POUNDS.

MATERIAL: ALLOY STEEL, AMS-6304, AMS-6485, AND AMS-6487.
HARDNESS: ROCKWELL C48 MAX.
SURFACE TEXTURE: BEARING SURFACE 125 IN ACCORDANCE WITH ANSI (ASA) B46.1-1962.
PLATING: CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2 DRY FILM LUBRICATED NUTS IN ACCORDANCE WITH QQ-P-416, TYPE AND CLASS OPTIONAL, IF THE NUTS WILL MEET THE SALT SPRAY REQUIREMENTS OF QQ-P-416, TYPE II.
LUBRICANT: LUBRICANT APPROVED IN ACCORDANCE WITH PROCUREMENT SPECIFICATION. LUBRICANTS, EXCEPT DRY FILM LUBRICANTS, SHALL BE SOLUBLE IN THE CLEANER SPECIFIED IN THE PROCUREMENT SPECIFICATION. FOR USAF APPLICATIONS, NUTS TREATED WITH DRY FILM LUBRICANTS SHALL NOT BE UTILIZED IN INTEGRAL FUEL TANKS.

THREADS: MIL-S-8879 BEFORE LUBRICATION.
WRENCHING ELEMENT: PER MS33787, DRIVERS PER MIL-W-8982.
PERPENDICULARITY: BEARING SURFACE SHALL BE NORMAL WITH PITCH DIAMETER OF THREAD WITHIN X WHEN CHECKED IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.
DIMENSIONS IN INCHES: DIMENSIONS APPLY BEFORE LUBRICATION
DESIGN AND USAGE INFORMATION: THESE NUTS ARE DESIGNED TO BE USED WITH MS21134 BOLTS AND MS21206 WASHERS.
EXAMPLE OF PART NUMBERS:
MS21133-04 .2500-28 NUT, CADMIUM PLATED, SOLUBLE LUBRICANT.
MS21133-L04 .2500-28 NUT, CADMIUM PLATED, DRY FILM LUBRICATED.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCE DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

APPROVED 28 JAN 1972 REVISED

| | | |
|---|---|---|
| P.A. NAVY - AS<br>Other Cust<br>USAF-11<br>ARMY-AV | TITLE<br>NUT, SELF-LOCKING, STEEL, 180 KSI Ftu, 450°F, FLANGED, MS33787 WRENCHING ELEMENT | MILITARY STANDARD<br>MS21133 |
| PROCUREMENT SPECIFICATION<br>MIL-N-8985 | SUPERSEDES: | SHEET 1 OF 1 |

DD FORM 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

U.S. GOVERNMENT PRINTING OFFICE: 1973-714-542/3635

"Review/user information is current as of the date of this document. For future coordination of changes to this document, draft changes should be based on the information in the current DODISS."

This military standard is mandatory for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

REVIEWER SYMBOLS:

USER SYMBOLS:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:27Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MS21133 NUT, SELF-LOCKING, STEEL, 180 KSI Ftu, 450°F, FLANGED, MS33787 WRENCHING ELEMENT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T14:27Z
Check the source to verify that this is the current version before use.